THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 7: **Particulate Nature of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Water kept in an earthen pot remains cool because: 1

(a) mud is cold
(b) sun heats it
(c) water is freezing
(d) water evaporates through tiny pores of the pot

2. Which state of matter has neither definite shape nor definite volume? 1

(a) ice **(b)** solid **(c)** gas **(d)** liquid

3. Which substance sublimes? 1

(a) salt **(b)** sugar **(c)** camphor **(d)** iron

4. A gas can be liquefied by: 1

(a) expanding
(b) decreasing pressure only
(c) heating
(d) increasing pressure and lowering temperature

5. Which of the following can be compressed easily? 1

(a) wood **(b)** solids **(c)** liquids **(d)** gases

6. During melting, the temperature of a pure solid: 1

(a) keeps rising **(b)** keeps falling **(c)** remains constant **(d)** becomes zero

7. LPG is stored in cylinders as: 1

(a) solid **(b)** liquid under high pressure
(c) plasma **(d)** powder

8. The forces of attraction between particles are strongest in: 1

(a) liquids **(b)** gases **(c)** same in all **(d)** solids

9. The heat energy absorbed during change of state without temperature rise is called: 1

(a) kinetic heat **(b)** latent heat **(c)** friction **(d)** specific heat

10. Dry ice is solid: 1

(a) carbon dioxide **(b)** nitrogen **(c)** water **(d)** oxygen

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SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why are gases easily compressed? 2

12. Why can solids not be compressed easily? 2

13. Why does the temperature stay constant during melting? 2

14. How does LPG get stored in a small cylinder? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. What are the factors that affect the rate of evaporation? Name three. 3

16. Give three examples of diffusion in daily life. 3

17. What happens to the particles of a liquid when it is heated? Explain boiling. 3

OR

Why does a desert cooler work better in dry weather? 3

18. Differentiate between melting point and boiling point. 3

OR

State three differences between the three states of matter on the basis of compressibility, density and shape. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain why the temperature remains constant during melting and boiling using the idea of latent heat and particles. 5

OR

Describe sublimation with three examples and explain how it can be used to separate a mixture of salt and camphor. 5

20. Explain how change in pressure and temperature can change the state of a substance. Give examples with LPG and dry ice. 5

OR

Why does water vapour from a boiling kettle appear as a white cloud only a little distance from the spout? Explain with particles. 5

END OF QUESTION PAPER — Total: 40 Marks

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